Machine-Learning Design of Broadband Rayleigh-Wave Carpet Cloaks

under Microstructure Realisability Constraints

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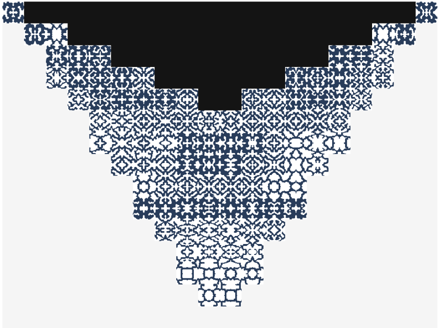
Outline

1. The dream: a cloak for surface waves
2. The catch: analytical cloaks are unbuildable
3. Idea: *let a network find the material*
4. Four ML routes to a real structure
   - Neural-field material optimisation
   - Unit-cell inverse design
   - Guided-diffusion generation
   - Latent-space search
5. Results, and how much survives fabrication



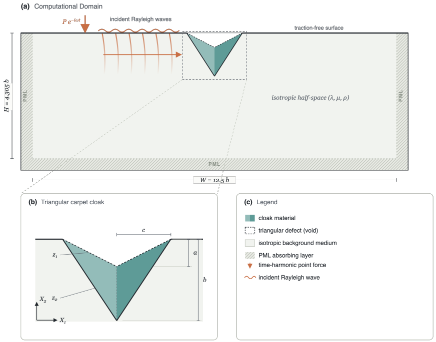
A cloak made of *one* material, patterned cell by cell.

Hiding from a surface wave

The goal

- A **Rayleigh wave** rolls along the free surface of a half-space. A surface *inclusion* scatters it and sensibly affects the displacement field beyond the defect.
- A **carpet cloak** is a coating that makes the surface wave behave as if the inclusion were not there.
- Applications: seismic protection, surface-acoustic-wave devices, non-destructive testing.

➤ **Takeaway.** We want a coating, buildable from *ordinary materials*, that restores the surface wavefield beyond the defect.



Half-space, PML boundaries, a time-harmonic point source, and the triangular cloak over the inclusion.

Transformation elasticity: the exact cloak

In the homogeneous reference (virtual) half-space $X = (X_1, X_2)$, the in-plane displacement $U = (U_1, U_2)$ obeys the Navier elastodynamic equation

$$\nabla_X \cdot (C : \nabla_X U) = \rho\, U_{tt},$$

A coordinate map $x = \Xi(X)$, $F = \nabla_X x$, $J = \det F$, and the gauge $U(\Xi(X)) = u(x)$ carry it to $\nabla_x \cdot (c^{eff} : \nabla_x u) = \rho^{eff} u_{tt}$ with the **Brun–Guenneau–Movchan** push-forward:

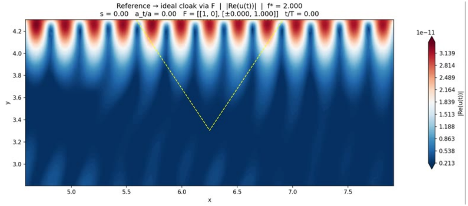
$$c^{eff}_{ijkl} = J^{-1}\, C_{IjKl}\, F_{iI}\, F_{kK}, \qquad \rho^{eff} = \rho\, J^{-1}.$$

The obstruction

The transformed tensor keeps the *major* symmetry but breaks the *minor* ones,

$$c^{eff}_{ijkl} = c^{eff}_{klij}, \qquad c^{eff}_{ijkl} \neq c^{eff}_{jikl}.$$

Hence the ideal cloak is a **polar (Cosserat)** medium with a non-symmetric stress.



The map Ξ morphing the flat reference half-space into the notched one.

ML methods for realisable cloaks

Objective

Cast carpet-cloak design as a **PDE-constrained optimisation** solved by a coordinate **neural field** through a **differentiable finite-element** wave solver, and carry the design all the way to a fabricable single-material microstructure.

1. **Neural-field material optimisation** – the main pipeline (Sec. 3).
2. **Unit-cell inverse design** with a neural field.
3. **Guided-diffusion** generation of unit cells.
4. **Latent-space search** over a learned design manifold.

Unconstrained design $\longrightarrow \eta = 0.99$ (near-ideal)

Best *fabricable* design $\longrightarrow \approx 97\%$ of ideal

First microstructured Rayleigh-wave cloak from a common material.

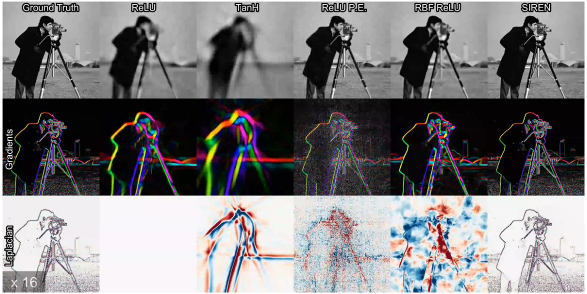
What is a neural field?

A small MLP that maps a **coordinate** to a **field value**,

$$\Phi_\theta : (x, y) \mapsto \text{field},$$

trained until it *is* the field. The weights θ, not a pixel grid, are the degrees of freedom.

- **Resolution-free**: query anywhere, e.g. at FEM quadrature points.
- **Compact**: $\mathcal{O}(10^3)$ weights for a whole material field.
- **Differentiable**: $\partial(\text{field})/\partial\theta$ in closed form.
- **Smooth by construction**: the architecture, not a penalty, sets the bandwidth.



Same target, same budget, different activation/encoding (Sitzmann et al., 2020). All fit the image (top); only some reproduce its **gradients** and **Laplacian** – plain ReLU has none.

Elastic cloaking optimization formulation

Design class: orthotropic Cauchy (4CM)

Square D_2-symmetric cells force $C_{16} = C_{26} = 0$:

$$C_{4CM} = \begin{bmatrix} C_{11} & C_{12} & 0 \\ C_{12} & C_{22} & 0 \\ 0 & 0 & C_{66} \end{bmatrix} \succ 0, \quad (+\rho).$$

Four moduli + density *per cell* are the design variables.

Objective: phase-invariant magnitude error

$$\mathcal{L}_{cloak}(\theta) = \frac{1}{|\mathcal{B}|} \int_{\mathcal{B}} \left(\frac{|u_{cloak}|}{|u_{ref}|} - 1 \right)^2 dA$$

over a near-surface strip $\mathcal{B}$ downstream of the cloak.

Metric: the **cloak ratio** $\eta = \dfrac{\langle |u| \rangle}{\langle |u_{ref}| \rangle} \to 1$ (perfect cloak).

Methodology: one differentiable pipeline

❶ Represent

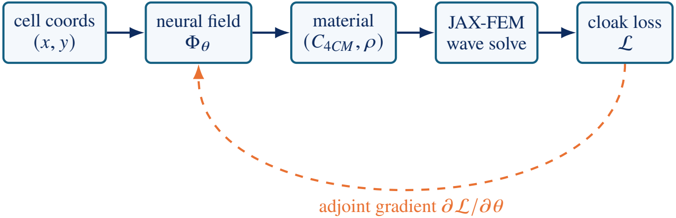
- A coordinate MLP Φ_θ *is* the material field.
- Outputs (C_{4CM}, ρ) per cell.
- Smoothness set by Fourier features.

❷ Solve & differentiate

- Frequency-domain JAX-FEM wave solve.
- Gradients via the implicit **adjoint** of the linear solve.
- End-to-end $\partial\mathcal{L}/\partial\theta$.

❸ Realise

- Project onto a **realisability manifold**.
- Fill each cell with a solid/void microstructure.
- Inverse design / diffusion / latent search.



❶ Method 1 – Neural-field material optimisation

Reparameterise the material field by a coordinate MLP

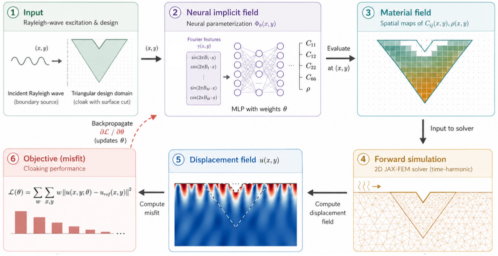
$$\Phi_\theta : (x, y) \mapsto (C_{4CM}(x, y),\ \rho(x, y)),$$

applied as a **multiplicative residual** around a physical prior:

$$C_{4CM} = C^{(0)}_{4CM} \odot \left(1 + \epsilon\, \Phi^{(C)}_\theta\right).$$

- tanh MLP, 4 layers, Fourier positional encoding.
- Relative perturbation is scale-uniform across moduli.
- Smoothness is *built in* – no neighbour penalty needed.

Neural reparameterization of a 2D Rayleigh-wave cloak



Coordinates $\longrightarrow$ Fourier encoding $\longrightarrow$ MLP $\longrightarrow$ (C_{4CM}, ρ) $\longrightarrow$ FEM $\longrightarrow$ loss, differentiated back to θ.

➤ **Takeaway.** The network is the material. Design = training its weights against a wave-physics loss.

❶ Admissible *by construction*: the decoding map

Raw outputs $r = (r_{11}, r_{22}, r_{66}, r_{12})$ are decoded so every design is a valid Cauchy tensor – no penalty policing after the fact.

Positivity

$$C_{ii} = C_{ii}^{(0)} e^{\epsilon r_{ii}}$$

Log-space $\Rightarrow C_{ii} > 0$ for any real r_{ii}.

Bounded anisotropy

$$g_b = \tfrac{1}{2} \log R \, \tanh\!\left(\frac{g}{\frac{1}{2} \log R} \right)$$

confines $C_{11}/C_{22} \in [1/R, R]$, $R = 30$.

PD coupling

$$C_{12} = \rho_c \sqrt{C_{11} C_{22}}$$

$\rho_c = \kappa \tanh(r_{12} + \phi_0)$, $\kappa = 0.99$
$\Rightarrow \det > 0$.

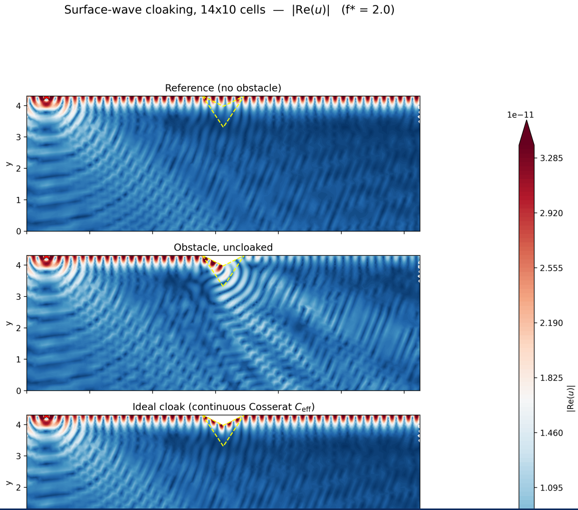
➤ **Takeaway.** A smooth, everywhere-differentiable surjection onto the set of positive-definite, bounded-anisotropy tensors: the optimiser can never leave the realisable class.

Results I – one material already beats symmetrisation

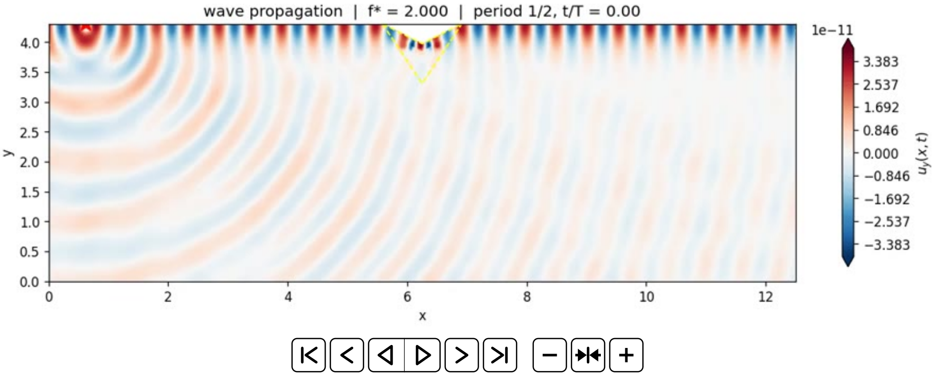
Cloak ratio η at $f^\star = 2.0$

Configuration	η
Obstacle (no cloak)	0.37
Symmetrised ideal (Chatz. 2023)	0.63
Single optimised 4CM	0.86
2×2 (four materials)	0.96
14×10 **grid (46 cells)**	0.99
Ideal continuous polar c^{eff}	0.999

Depth beats width: subdividing in depth helps far more than along propagation.



Results I – the cloak in motion



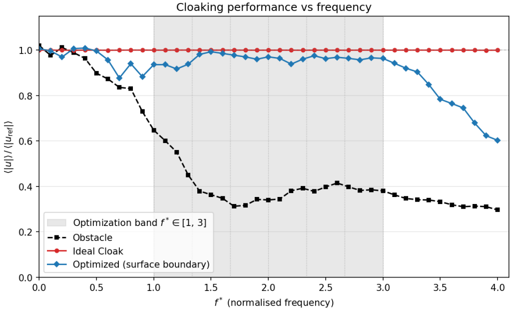
Vertical displacement u_y at $f^\star = 2.0$; the surface wave passes the cloaked notch and re-forms downstream.

➤ **Takeaway.** The wavefront leaves the cloak intact – no shadow, no reflected front.

Train a single neural field against a band of frequencies at once:

$$\mathcal{L}_{band}(\theta) = \frac{1}{\sum_f w_f} \sum_{f \in \mathcal{S}} w_f \, \mathcal{L}_{cloak}(\theta; f).$$

- Band $f^\star \in [1, 3]$, 7 frequencies, equal weights.
- Optional min–max variant flattens the worst case.
- Cost scales linearly in $|\mathcal{S}|$; frequencies solved in parallel.
- Floquet–Bloch check: propagating modes, *no* resonant band-gap trickery.



$\eta(f)$: broadband design (blue) holds ≈ 0.96 across the shaded training band; degrades gracefully outside it.

➤ **Takeaway.** The single-frequency narrowness was a limit of the *objective*, not the material class.

From numbers to structures: the realisability problem

Mechanical Acoustic Thermal

The gap

Optimised cells carry a homogenised (C_{4CM}, ρ) that need not correspond to any manufacturable object. We must *realise* each cell as a real two-phase (solid/void) microstructure.

1. **Dataset:** $N \sim 10^6$ binary 50×50 cells (cellular-automaton growth).
2. **Homogenise:** periodic FEM $\longrightarrow (C_{4CM}, \rho_{eff})$; D_2 symmetry $\Rightarrow$ exactly the 4CM form.
3. **Prior:** fit a GMM; keep a log-density super-level set as the realisable manifold $\mathcal{M}$.
4. **Constrain:** add the manifold penalty during optimisation.

Manifold penalty

$$\mathcal{L}_{GMM} = \sum_c \max\big(0,\ \tau - \log p_{GMM}(C^c_{4CM}, \rho_c)\big)$$

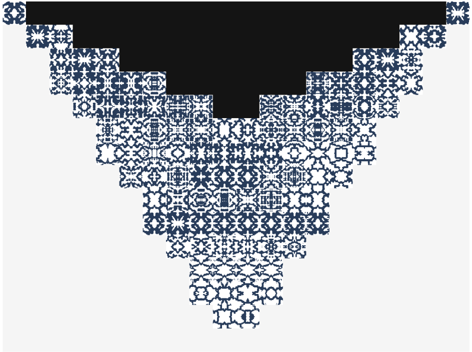
pulls every cell into $\mathcal{M}$ without pinning it to one catalogue entry.

Given a per-cell target $y = (C_{4CM}, \rho)$, parameterise a *continuous density field* over the 50×50 cell, $\hat{\rho}(x, y) = g_\phi(x, y) \in [0, 1]$, and minimise

$$\min_{\phi} \left\| \mathcal{H}(\hat{\rho}_\phi) - y \right\|^2 + \lambda \mathcal{R}_{conn},$$

where $\mathcal{H}$ is the differentiable periodic-FEM homogenisation operator.

- Soft-to-hard schedule drives $\hat{\rho} \to \{0, 1\}$.
- Connectivity penalty $\mathcal{R}_{conn}$ kills floating islands/voids.
- **Deterministic**: one locally optimal cell per target.



Inverse-design realisation: one cement phase, cell-varying void pattern tiled across the carpet.

Method 3 – Guided-diffusion generation

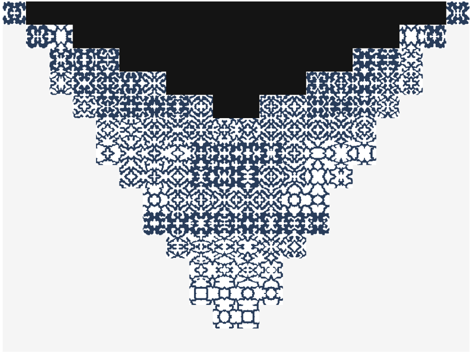
Train a **conditional denoising diffusion** model on the catalogue,

$$\min_{\theta}\ \mathbb{E}_{t,x_0,\boldsymbol{\epsilon}}\left\|\boldsymbol{\epsilon}-\boldsymbol{\epsilon}_\theta(x_t,t,y)\right\|^2,\quad y=(C_{4CM},\rho),$$

then sample the reverse process $p_\theta(x_{t-1}\,|\,x_t,y)$. Because sampling is stochastic, draw N candidates and keep the closest:

$$x^\star=\arg\min_{n=1..N}\left\|\mathcal{H}(x^{(n)})-y\right\|\quad(\textbf{best-of-}N).$$

- Prior keeps every sample *on* the manufacturable manifold.
- Explores a diverse set of realisations per target.



Best-of-N diffusion realisation of the same target field – visibly different, equally manufacturable patterns.

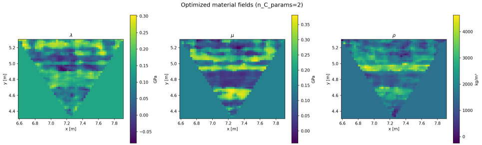
❹ Method 4 – Search in a learned latent space

Direct pixel optimisation is ill-conditioned ($\sim 10^4$ DoFs, bad minima). Train an **autoencoder** on $(C, \rho, f_\star, \mathcal{L})$ with a *frequency-separated* latent

$$z = z_{sh} + z_f(f_\star),$$

(geometry vs. per-frequency offset), a performance head, and a within-frequency contrastive ranking on z_{sh}. Then **freeze** the AE and optimise the *code* z:

$$\min_z \ \alpha \max_i \mathcal{L}(f_i; z) + (1-\alpha)\frac{1}{N}\sum_i \mathcal{L}(f_i; z), \quad \alpha : 1 \to 0.$$



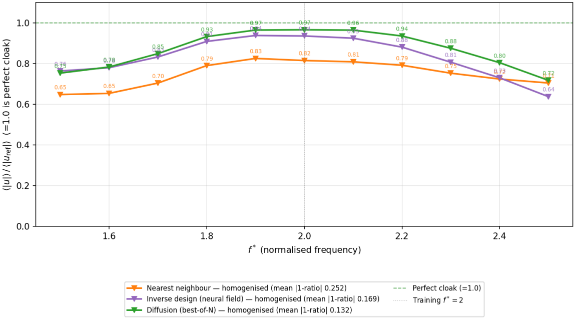
AE latent decodes to structured, *manufacturable* motifs. Cf. `frequency_sweep_vis.png`: AE $\eta \approx 0.965$, neural-field ≈ 0.985.

➤ **Takeaway.** The latent search inherits the training distribution's manufacturable priors – smoother layouts, fewer iterations than raw topology optimisation.

Results III – how much survives fabrication?

Realising the 20×15 cloak (100 cells), $f^\star = 2.0$

Realisation of the 100 cells	η
Continuous 4CM optimum	0.99
Nearest-neighbour snap	0.82
Inverse design (neural field)	0.94
Diffusion (best-of-*N*)	0.97
Ideal continuous polar c^{eff}	0.999



$\eta(f)$ across the band: diffusion (green) stays closest to the perfect-cloak line; smallest mean distortion.

➤ **Takeaway.** Better per-cell matching ⟹ better cloak: best-of-N diffusion recovers **97%** of the ideal, from fabricable single-material cells.

Conclusions

- A single **differentiable pipeline** – coordinate neural field + JAX-FEM adjoint – designs Rayleigh-wave carpet cloaks inside the realisable Cauchy class.
- **Near-perfect** single-frequency ($\eta = 0.99$) and **broadband** (≈ 0.96 over an octave) cloaking, beating closed-form symmetrisation.
- Four ML routes to a real structure – neural-field optimisation, inverse design, guided diffusion, latent search – with **diffusion recovering** $\approx 97\%$ of ideal performance from a *single* constituent.

Future work: a differentiable generative manifold queried *during* optimisation; multiscale generation with connected neighbours on a body-fitted mesh; 3-D cloaks; and polar/Cosserat microstructures to close the broken-minor-symmetry gap.

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(Placeholder entries – replace with the paper's `cas-refs.bib` *for the final deck.)*

Thank You

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